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# Angelfish Pattern Dynamics and the Evidence for a Reaction-Diffusion Mechanism: Comment on Kondo and Asai (1995)

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## Abstract

This comment was submitted to *Nature* in 1995 in response to Shigeru Kondo and Rihito Asai, "A reaction-diffusion wave on the skin of the marine angelfish *Pomacanthus*" (*Nature* 376, 765-768). It argues that similarity of pattern dynamics does not by itself establish a Turing reaction-diffusion mechanism. Closely corresponding dynamics can arise in models with different underlying mechanisms, illustrated here with a model of ocular-dominance pattern formation. The comment concludes that a stronger demonstration of a Turing mechanism would require identification of the relevant morphogens together with characterization of their reaction kinetics and transport properties.

## Keywords

reaction-diffusion, Turing mechanism, biological pattern formation, stripe formation, *Pomacanthus imperator*, emperor angelfish, nonlinear pattern selection, ocular dominance, Swindale model, Kondo and Asai

## Provenance

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Sir-

Kondo and Asai[1] are to be congratulated for describing a beautiful illustration of the dynamics of biological pattern formation in the marine angelfish *Pomocanthus*. Nature magazine, however, has perhaps overstated the case that a Turing mechanism underlies the angelfish stripes with the claim, found in the legend to the cover photograph, that this is "the first clear example of Turing's reaction-diffusion wave in a biological system."

Pattern formation phenomena identical to those reported by Kondo and Asai may be observed in other systems with mechanisms that do not involve reaction-diffusion. For example, the accompanying figure, analogous to Kondo and Asai's figure 3, shows joint switching in the time evolution of Swindale's model[2,3] for the genesis of ocular dominance domains in the developing primary visual cortex. Such models for cortical development may be derived from a Hebbian synaptic plasticity mechanism[4]. This model also fully replicates the dynamics reported in figure 1 (intercalation of new stripes with growth) and figure 2 (slow movement of pattern branching points).

Analytical work[5] reveals the origin of this resemblance of the dynamical behaviour of the distinct models. It lies both in the linear properties of the models, which

determine the wavelength of pattern formation, and in the nature of the nonlinear coupling between the different spatial modes that superpose to form a particular two-dimensional pattern. Many other patterning systems give rise to similar couplings between pattern modes, and dynamics that correspond closely to those seen on the angelfish skin.

Similarity of pattern dynamics is no more conclusive evidence that a Turing mechanism is at work than similarity of the patterns themselves. The most that can be concluded from this similarity is that whatever the form of the physiological mechanism for angelfish skin patterning, it probably involves a locally self-enhancing process coupled with lateral inhibition and an appropriate form of nonlinearity. For stripe selection, a predominance of the cubic nonlinearity[5], or Meinhardt's "autocatalysis saturation"[6] is sufficient.

Computer simulation plays an indispensable role in understanding the non-intuitive behaviour of complex mathematical models but cannot be taken alone as proof of mechanism. A solid demonstration that a Turing mechanism underlies patterning would also require identification of the morphogens and characterization of their reaction kinetics and transport properties. An extensive review of such evidence, as yet incomplete, but suggesting that mechanisms like the one first proposed by Alan Turing[7] do play a critical role in biological development, may be found in Harrison's recent monograph[8] on the subject.

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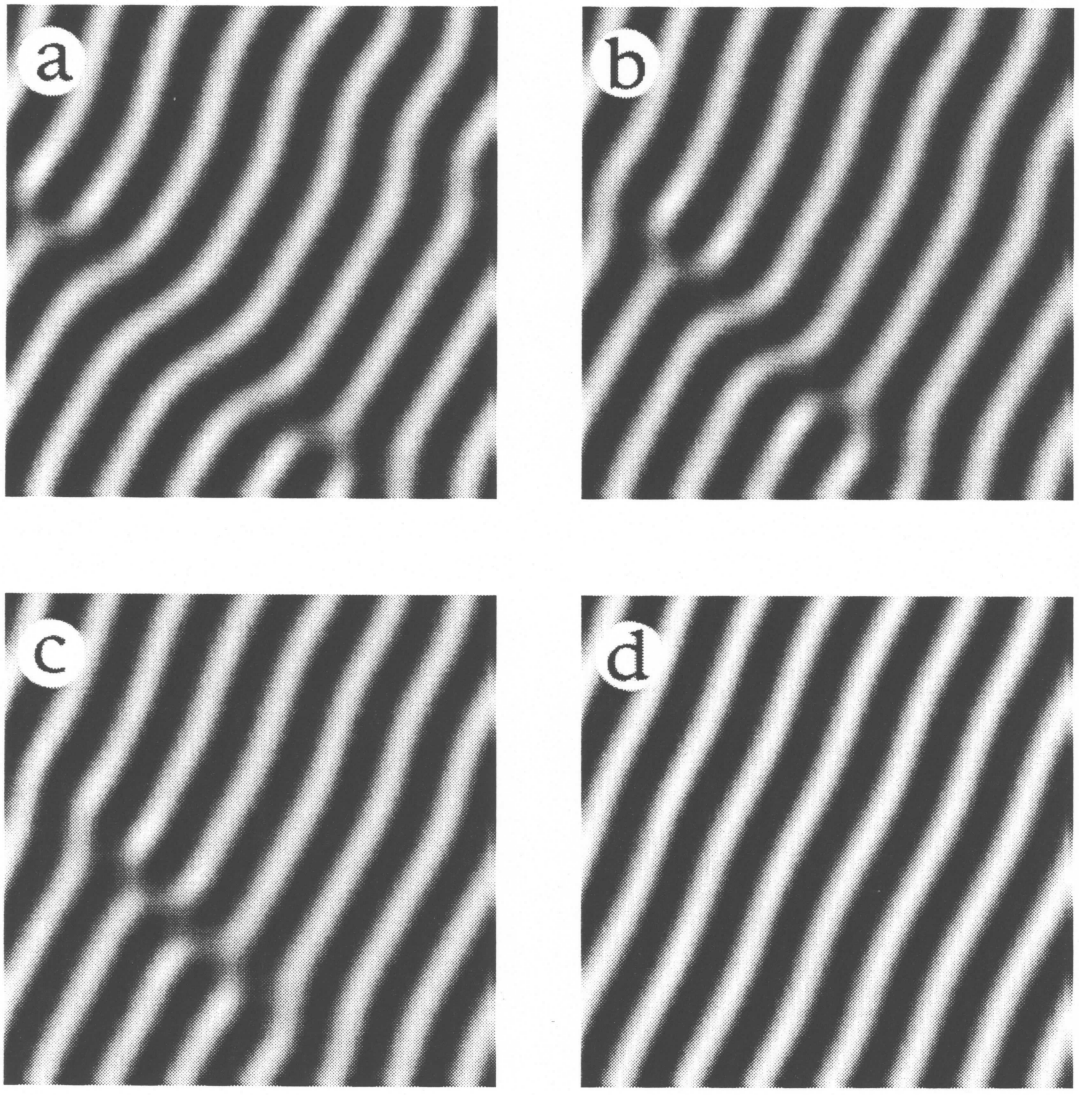
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# Figure Legend

Pattern joint switching observed with a modified form of Swindale's equation for the development of ocular dominance domains in the cerebral cortex. The pattern defects move laterally with respect to the stripes until they meet and annihilate one another. The figure shows the time evolution of the integro-differential equation:

$$\frac{\partial a}{\partial t} = f(a) + \int w(r - r')a(r')d^2r' \text{ with } w(r) = A\, e^{-r^2/r_A} - B\, e^{-r^2/r_B},$$

where $f(a) = \lambda a + Qa^2 + Ca^3$, and the integral is evaluated over the patterning domain. Greyscaled density plots of $a(r)$ on a 64x64 domain with wrap-around boundaries. In the initial condition (not shown) the values of $a(r)$ at each site of the latice were chosen from a uniform distribution on $(-0.1, 0.1)$ using a pseudo-random number generator. Relative to an intermediate stage of the computation in (a), the pattern is shown at (b) 3000 iterations, (c) 4000 iterations, and (d) 5000 iterations later. The equation was solved numerically on the Caltech/JPL Cray using a second-order Runge-Kutta method and Fourier transformation to evaluate the integral at each iteration. The parameters used for this calculation were: $\lambda = -1.0$, $Q = 0.0$, $C = -1.0$, $A = 0.2$, $B = 0.1$, $r_A = 5.0$, $r_B = 20.0$, and the timestep 0.05.


a
b
c
d